

Degree forty-four in the closure of seven-product polynomial evaluations

A negative resolution of MF-14

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Abstract

Starting from $1, x$ and allowing free complex linear combinations, we prove that the Zariski closure of the polynomials computable with seven products contains every polynomial of degree at most 44. This refutes the equality to 42 in MF-14. A simultaneous four-product degeneration retains two intermediate polynomials as well as an arbitrary polynomial of degree at most 12. Three further products give a coefficient map with a 45×45 integer Jacobian of determinant 256. The published dimension theorem of Jarlebring and Lorentzon additionally implies $44 \leq d_7 \leq 47$.

Provenance and review. The construction and earlier computational checks were developed with substantial ChatGPT assistance in the conversation supplied by the author [3]. This standalone exposition uses the polynomial-determinant version of the border argument in that conversation. Separate Codex agents reviewed the border lemma and the continuation/certificate; their reports and independently written exact verifiers accompany this manuscript. These are informal AI-agent reviews, not external human peer review or formal verification.

1 Statement and computational model

Let $\mathcal{P}_m$ be the outputs obtainable from $1, x$ using arbitrary complex linear combinations of previously available polynomials and at most m products of two previously available polynomials. All intermediate polynomials have degree at most 2^m . Put

$$X_7 = \overline{\mathcal{P}_7}^Z \subseteq \mathbb{C}[x]_{\leq 128}, \quad d_7 = \max\{d : \mathbb{C}[x]_{\leq d} \subseteq X_7\}.$$

All closures below are Zariski closures in the indicated full coefficient spaces. In particular, approximating outputs are allowed to have degree greater than the degree of their limits. No coefficient projection is an allowed computational operation.

Theorem 1. *One has $\mathbb{C}[x]_{\leq 44} \subseteq X_7$. Consequently the equality $d_7 = 42$ asked in MF-14, and in Conjecture 13 of [1], is false.*

The previously audited degree-42 coverage result of Colbrook [2] remains valid. The new proof uses degenerating intermediate computations. It does not assert exact representation of every degree-44 polynomial, real Euclidean density, or stable coefficient recovery.

2 Simultaneous availability after four products

Let $\mathcal{A}_4 \subseteq (\mathbb{C}[x]_{\leq 16})^4$ be the ordered quadruples simultaneously available after at most four products, including arbitrary free linear combinations of the computed polynomials. Set $\mathcal{B}_4 = \overline{\mathcal{A}_4}^Z$.

Lemma 2. *For all $\alpha, \beta \in \mathbb{C}$ and $P \in \mathbb{C}[x]_{\leq 12}$,*

$$(x^2, x^4 + \alpha x^3, x^5 + \beta x^3, P) \in \mathcal{B}_4.$$

Proof. Fix α, β , put $\eta = \beta + \alpha^2$, and initially fix an arbitrary $\gamma \in \mathbb{C}$. For nonzero s tending to zero, put

$$Q = x^4 + \alpha x^3, \quad R_s = s^2 Q^2 + \gamma s x^2 Q + xQ + \eta x^3. \quad (1)$$

Three products. The first two products give x^2 and $Q = x^2(x^2 + \alpha x)$. For $a = s^2$, $b = \gamma s$, define

$$g = b - a, \quad \delta = b - 2a, \quad \sigma = \frac{\eta - 1 + \alpha g}{\delta}, \quad \rho = 1 - a\sigma.$$

Direct multiplication gives the identity

$$R_s = (aQ + \rho x + gx^2)(Q + \sigma x + x^2) - gQ - \rho\sigma x^2. \quad (2)$$

Every factor uses only already available polynomials, so this costs one additional product. The denominator $s(\gamma - 2s)$ is nonzero for all sufficiently small nonzero s , also when $\gamma = 0$. Moreover,

$$R_s - \alpha Q \longrightarrow R_0 - \alpha Q = x^5 + \beta x^3. \quad (3)$$

The fourth-product image is dense in its product span. Let $V_s = \text{span}\{1, x, x^2, Q, R_s\}$ and $W_s = \text{span}\{uv : u, v \in V_s\}$. Products from $U = \text{span}\{1, x, x^2, Q\}$ span exactly $\mathbb{C}[x]_{\leq 6} + \text{span}\{Q^2\}$: obtain x^3, x^4 directly and then $x^5 = xQ - \alpha x^4$, $x^6 = x^2Q - \alpha x^5$. Since R_s already lies in that span,

$$W_s = \mathbb{C}[x]_{\leq 6} + \text{span}\{Q^2, xR_s, x^2R_s, QR_s, R_s^2\}. \quad (4)$$

The five extra polynomials have respective degrees 8, 9, 10, 12, 16 and nonzero leading coefficients for $s \neq 0$. Thus $\dim W_s = 12$.

Consider $\Psi_s : V_s^3 \rightarrow W_s$, $(u, v, w) \mapsto uv + w$. At $u = R_s$, $v = Q + \lambda x^2$, its differential includes these twelve columns:

$$1, x, x^2, Q, R_s, x(Q + \lambda x^2), x^2(Q + \lambda x^2), Q(Q + \lambda x^2), xR_s, x^2R_s, QR_s, R_s^2. \quad (5)$$

In coefficient rows 0, 1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 16, their determinant is

$$a^5(\lambda - \eta - \alpha\lambda(b - a\lambda)). \quad (6)$$

This polynomial identity follows by coefficient expansion; the accompanying exact symbolic verifier also recomputes it. Choose $\lambda = \eta + 1$. The determinant becomes

$$s^{10}[1 - \alpha(\eta + 1)(\gamma s - (\eta + 1)s^2)],$$

which is nonzero for sufficiently small nonzero s , for every fixed choice of α, η, γ . Hence Ψ_s has full differential rank. Restricting to twelve suitable affine directions and applying the complex inverse function theorem shows that its image contains a Euclidean-open subset of W_s , so it is Zariski dense in W_s . All actual outputs $uv + w$ cost one further product and retain the previous intermediates. Closedness therefore gives

$$(x^2, Q, R_s - \alpha Q, p) \in \mathcal{B}_4 \quad (p \in W_s). \quad (7)$$

This is a closure argument, not an identification of a sum of products with one actual product.

An explicit polynomial degeneration. Set $(f_1, f_2, f_3, f_4, f_5) = (x^2 R_s, Q^2, Q R_s, R_s^2, x R_s)$ and define

$$E_s(x) = \det \begin{pmatrix} [x^7]f_1 & \cdots & [x^7]f_5 \\ [x^8]f_1 & \cdots & [x^8]f_5 \\ [x^9]f_1 & \cdots & [x^9]f_5 \\ [x^{10}]f_1 & \cdots & [x^{10}]f_5 \\ f_1(x) & \cdots & f_5(x) \end{pmatrix}. \quad (8)$$

Expansion along the last row gives $E_s \in W_s$ and repeated rows give $[x^j]E_s = 0$ for $7 \leq j \leq 10$. Direct polynomial expansion gives

$$[x^{11}]E_s = (4\gamma^2 - 3\alpha)s^4 + O(s^5), \quad (9)$$

$$[x^{12}]E_s = -s^4 + O(s^5), \quad (10)$$

while the coefficients in degrees 13, 14, 15, 16 are divisible by s^5, s^6, s^7, s^7 , respectively. Here the statements are identities in $\mathbb{Z}[\alpha, \eta, \gamma, s]$, so the remainder notation denotes exact divisibility. The verifier expands the cofactors of E_s with integer polynomial arithmetic and checks every coefficient and divisibility. Appendix A supplies explicit identities for them. Degree is at most 16.

Write $\pi_{\leq 6}$ for coefficient truncation used only in this algebraic identity. Since $\mathbb{C}[x]_{\leq 6} \subseteq W_s$, the polynomial

$$Z_s = -s^{-4}(E_s - \pi_{\leq 6}E_s) \in W_s \quad (s \neq 0) \quad (11)$$

has coefficients polynomial in s , and satisfies

$$Z_0 = x^{12} + (3\alpha - 4\gamma^2)x^{11}. \quad (12)$$

The subtraction in (11) occurs inside W_s ; no truncation of an actual circuit output is being charged as free.

The eleven polynomials

$$1, x, \dots, x^6, \quad x^2 R_s, \quad Q^2, \quad Q R_s, \quad R_s^2$$

converge to a basis of $\mathbb{C}[x]_{\leq 10}$. Indeed $R_0 = x^5 + \alpha x^4 + \eta x^3$, and the four last limits are monic of degrees 7, 8, 9, 10; their coefficient matrix together with the first seven monomials is triangular with determinant one.

For $P = \sum_{j=0}^{12} p_j x^j$ with $p_{12} \neq 0$, choose a complex root

$$\gamma^2 = (3\alpha - p_{11}/p_{12})/4.$$

Then $P - p_{12}Z_0 \in \mathbb{C}[x]_{\leq 10}$. Express this difference in the limiting eleven-element basis and use the same constant coefficients at $s \neq 0$. Adding $p_{12}Z_s$ gives $p_s \in W_s$ with $p_s \rightarrow P$ in all coefficients through degree 16. Taking limits in (7), using (3) and Euclidean closedness of the algebraic set $\mathcal{B}_4$, proves the lemma for $p_{12} \neq 0$. For $p_{12} = 0$, apply that result to $P + \varepsilon x^{12}$ and take $\varepsilon \rightarrow 0$. $\square$

3 Three further products and an exact coefficient certificate

Take

$$Q = x^4 + \alpha x^3, \quad R = x^5 + \beta x^3, \quad P = x^{12} + \sum_{j \in J} \xi_j x^j, \quad J = (3, 6, 7, 8, 9, 10, 11).$$

Define $L_3 = (x, x^2, Q)$, $L_4 = (x, x^2, Q, R)$, $L_5 = (x, x^2, Q, R, P)$, and compute

$$F = (P + \mathbf{u} \cdot L_4)(R + \mathbf{v} \cdot L_3), \quad (13)$$

$$G = (F + \mathbf{a} \cdot L_5)(R + \mathbf{b} \cdot L_3), \quad (14)$$

$$H = (G + \mathbf{c} \cdot L_6)(G + \mathbf{d} \cdot L_6), \quad L_6 = (x, x^2, Q, R, P, F). \quad (15)$$

The final output is

$$p = \mathbf{z} \cdot (1, x, x^2, Q, R, P, F, G, H). \quad (16)$$

The degrees of F, G, H are at most 17, 22, 44 for all parameters. Order the parameters by

$$(\alpha, \beta, \xi_3, \xi_6, \xi_7, \xi_8, \xi_9, \xi_{10}, \xi_{11}, \mathbf{u}, \mathbf{v}, \mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}, \mathbf{z}), \quad (17)$$

with entries within each vector in their displayed order. The vector lengths are 4, 3, 5, 3, 6, 6, 9, and $\mathbf{z} = (z_0, \dots, z_8)$. There are $2 + 7 + 4 + 3 + 5 + 3 + 6 + 6 + 9 = 45$ parameters, defining a polynomial map $\Phi : \mathbb{C}^{45} \rightarrow \mathbb{C}[x]_{\leq 44}$ with integer coefficients.

Why its image lies in the full seven-product closure. Fix the continuation coefficients in (13)–(16). They define a polynomial map $T : (\mathbb{C}[x]_{\leq 16})^4 \rightarrow \mathbb{C}[x]_{\leq 128}$ using the four input entries in place of x^2, Q, R, P , while retaining the free input x . On arbitrary inputs its product degrees are at most 32, 48, 96, so its codomain is indeed the full degree-128 space. For every quadruple in $\mathcal{A}_4$, the output costs at most seven products. By continuity,

$$T(\mathcal{B}_4) \subseteq \overline{T(\mathcal{A}_4)}^Z \subseteq X_7.$$

For example the first inclusion follows by taking the inverse image of the closed set on its right. Lemma 2 therefore implies $\Phi(\mathbb{C}^{45}) \subseteq X_7$. All high-degree coefficients are retained in this argument and vanish by continuity at the limit.

A nonsingular integer Jacobian. Set every parameter to zero except

$$\alpha = b_2 = d_3 = d_5 = d_6 = z_8 = 1. \quad (18)$$

Then

$$Q = x^4 + x^3, \quad R = x^5, \quad P = x^{12}, \quad F = x^{17}, \quad G = x^{22} + x^{19}, \quad H = G(G + Q + P + F).$$

For an explicit derivative table at this point, set $B = R + x^2$, $K = 2G + Q + P + F$, $L = KB + G$. The Jacobian columns, viewed as polynomials, in the order (17), are

Parameter group	Derivative polynomials
α	Gx^3
β	$(LP + KF)x^3$
$\xi_j \ (j \in J)$	$(LR + G)x^j$
$\mathbf{u}$	$LR L_4$
$\mathbf{v}$	$LP L_3$
$\mathbf{a}$	$KB L_5$
$\mathbf{b}$	$KF L_3$
$\mathbf{c}$	$(G + Q + P + F)L_6$
$\mathbf{d}$	GL_6
$\mathbf{z}$	$(1, x, x^2, Q, R, P, F, G, H)$

With coefficient rows $0, \dots, 44$, exact elimination gives

$$\det D\Phi = 256 \neq 0. \quad (19)$$

The complete integer matrix is supplied with the independent verifier. It is reconstructed by simultaneous first-order polynomial arithmetic and compared entrywise with this derivative table. All coefficients are retained. The entries lie between 0 and 6; modular determinants are 1, 54, 256 modulo 3, 101, 1009, respectively. A separate whole-circuit interpolation check recovered from the original conversation is also supplied.

The complex inverse function theorem now shows that the image of Φ is Zariski dense in $\mathbb{C}[x]_{\leq 44}$. Since X_7 is closed and contains that image, it contains $\mathbb{C}[x]_{\leq 44}$. This proves Theorem 1.

4 An additional upper bound

Corollary 3. *Using Theorem 10 of Jarlebring and Lorentzon [1], one has $44 \leq d_7 \leq 47$.*

Proof. All seven-product computations can be parameterized by one affine space: at step k , multiply two arbitrary linear combinations of $1, x$ and the $k-1$ previous product outputs, and finally take an arbitrary linear combination of $1, x$ and the seven product outputs. Unused products can be padded by zero products. Thus X_7 is the closure of the image of an irreducible affine space, and is irreducible. The cited dimension theorem gives $\dim X_7 = 49$. If $\mathbb{C}[x]_{\leq 48} \subseteq X_7$, this closed irreducible subspace has the same dimension as X_7 , so it must equal X_7 . But seven successive squarings compute x^{128} , contradicting that equality. Therefore $d_7 \leq 47$, and the lower bound follows from Theorem 1. $\square$

Determining which of 44, 45, 46, 47 equals d_7 remains open in this contribution. That further question is not needed to refute the retained MF-14 equality. The lower bound and negative resolution do not depend on the external dimension theorem.

5 Reproduction and evidence scope

The accompanying submission README records exact commands, source hashes, machine-readable certificates and two separate review reports. The symbolic border verifier and the coefficient-map verifier use exact integer/rational arithmetic. Their checks validate the finite identities above; the analytic and algebraic closure arguments have been informally reviewed by separate agents. No Lean proof or kernel-checking claim is made.

A Explicit high-degree coefficient identities

Let c_i be the signed cofactor multiplying f_i in the expansion of (8) along its last row. The first, second and fifth polynomials f_i have degree at most ten, so only $c_3 Q R_s + c_4 R_s^2$ contributes above degree ten. Direct expansion of the 4×4 cofactors gives $c_3 = s^2 C(s)$ and $c_4 = s^3 D(s)$, where

$$\begin{aligned} C(s) = & -1 - 3\alpha\gamma s + (-2\alpha^3 + 9\alpha^2\gamma^2 - 6\alpha\eta - 2\eta\gamma^2)s^2 \\ & + (-4\alpha^4\gamma + 3\alpha^3\gamma^3 + 12\alpha^2\eta\gamma)s^3 \\ & + (-8\alpha^6 - 4\alpha^5\gamma^2 - 10\alpha^4\eta)s^4, \\ D(s) = & 2\gamma + (3\alpha^2 - \alpha\gamma^2 - \eta)s - 8\alpha^3\gamma s^2 + 9\alpha^5 s^3. \end{aligned}$$

Consequently the exact high-degree coefficients are

$$\begin{aligned}
[x^{11}]E_s &= s^4\{3\alpha C + D[2\gamma + (6\alpha^2 + 2\eta + 2\alpha\gamma^2)s + 2\alpha^3\gamma s^2]\}, \\
[x^{12}]E_s &= s^4\{C + sD[(6\alpha + \gamma^2) + 6\alpha^2\gamma s + \alpha^4 s^2]\}, \\
[x^{13}]E_s &= s^5 D(2 + 6\alpha\gamma s + 4\alpha^3 s^2), \\
[x^{14}]E_s &= s^6 D(2\gamma + 6\alpha^2 s), \\
[x^{15}]E_s &= 4\alpha s^7 D, \quad [x^{16}]E_s = s^7 D.
\end{aligned}$$

These formulas directly give (9)–(10) and all required divisibilities, including special choices where leading coefficients vanish. They are checked independently against the full determinant expansion by the border verifier.

References

- [1] E. Jarlebring and G. Lorentzon, *The polynomial set associated with a fixed number of matrix-matrix multiplications*, arXiv:2504.01500v3, Theorem 10 and Conjecture 13. <https://arxiv.org/html/2504.01500v3>.
- [2] M. J. Colbrook, MF-14 submission, 11 September 2026, Theorem 1 and equations (1)–(2). Archived in [the OpenProblemsInNLA manuscript archive](#).
- [3] Author-supplied ChatGPT conversation, accessed 17 September 2026. Degree-44 construction and subsequent independent-method validation within the same conversation. <https://chatgpt.com/share/6aaba6d9-3424-83eb-b278-21e3b917a83b>.